Application of GHZ State in Space-based FSO/QKD System

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Outline

1. Review of Previous Paper
2. New Proposal

1. Review of Previous Paper

Background: Security of 6 G Networks

The Emergence of 6G: Driving the need for secure group communications

- Simultaneous data exchange among massive distributed wireless users
- Require highly secure, low-latency group communication, wide-area

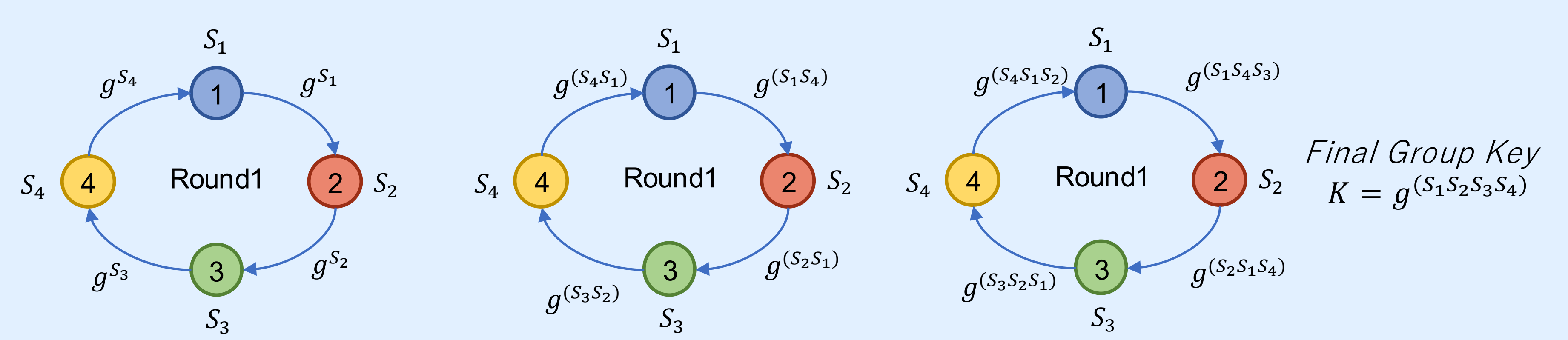
The Threat of Quantum Computers: Classical cryptography is vulnerable to future quantum computer attacks.

Scalable and efficient distribution of a **quantum-resistant group key** to support secure group communications.

Research Motivation: Conventional Group Key Protocol

Conventional Approach (e.g., ING Protocol)
Nodes form a logical ring, exchanging partial keys sequentially

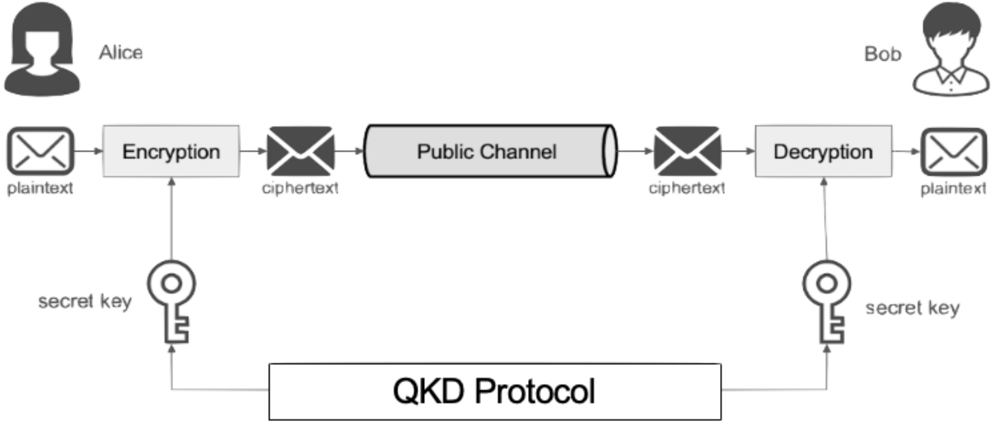
- Key Demerits (M – 1) sequential rounds — e.g., 99 rounds for M=100 users
- Computationally intensive exponentiation at every round
- Security relies on discrete-log hardness, broken by quantum computer attack



Quantum Key Distribution (QKD)

QKD is a secure communication technology to distribute secret keys

- Based on quantum physics.
- Guarantees unconditional security



Extension to Group Key Distribution:

- **N-GHZ-QKD** is specialized protocol for direct group key generation using multi-partite entanglement.
- Expressed as $|GHZ_N\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}\right)$

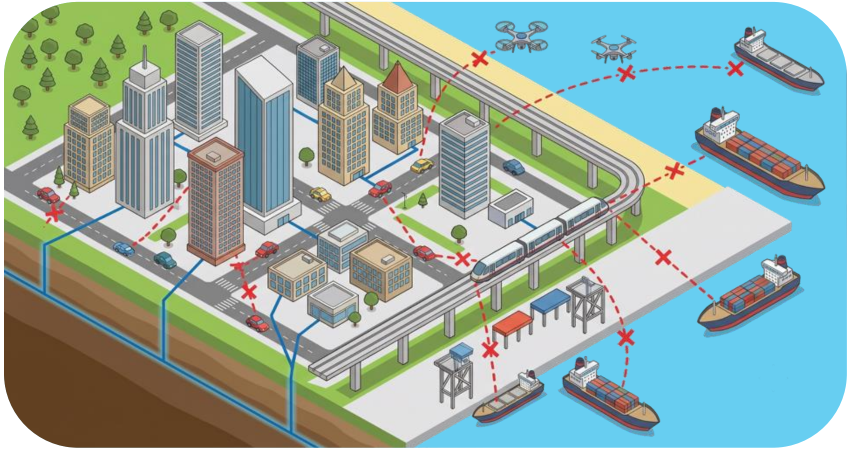
Research Motivation - Limitations

Current QKD Implementations:

- Rely on optical fiber, deployed between fixed points
- Impractical for massive, distributed wireless users
  → Fiber-based QKD does not scale to wide-area 6G networks

Current N-GHZ-QKD Limitations:

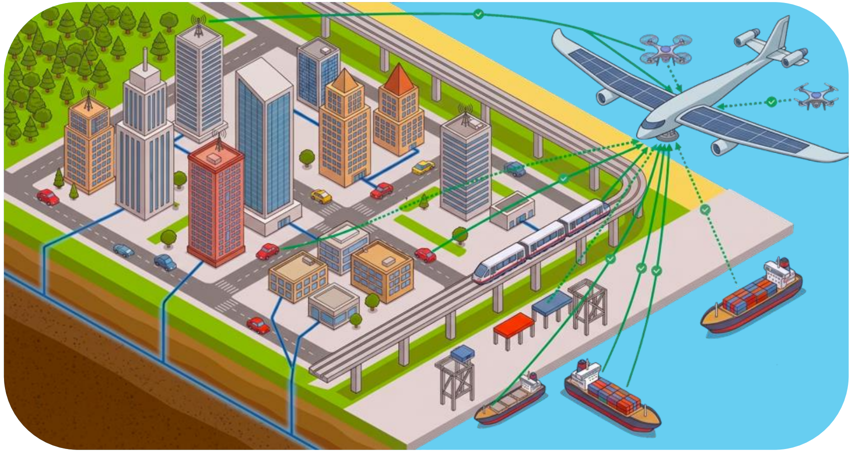
- Large-scale N-GHZ states can theoretically generate the group key for all users at once
- Generating/distributing such states remains experimental
  → Not yet practically feasible



Research Motivation - Possible Solution

High-Altitude Platform (HAP):

- Operates in the stratosphere (Approximately 20km)
- Wireless FSO links replace physical fiber



Small-scale GHZ-QKD with TDMA:

- Generating small-scale GHZ is practically achievable.
- Generate keys per small group via GHZ-QKD, then use them to encrypt and distribute the group key.

A scalable HAP-assisted N-GHZ-QKD architecture over FSO, combining small-scale GHZ-QKD with TDMA for wide-area group key distribution.

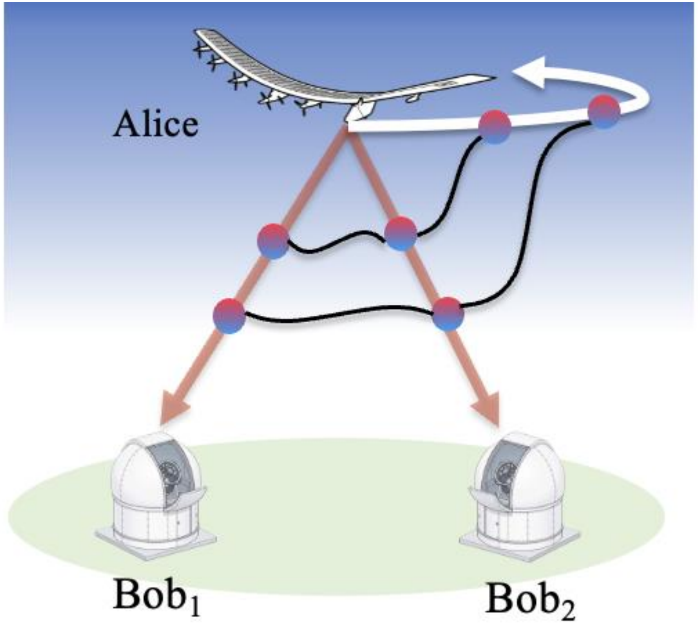
Research Goals

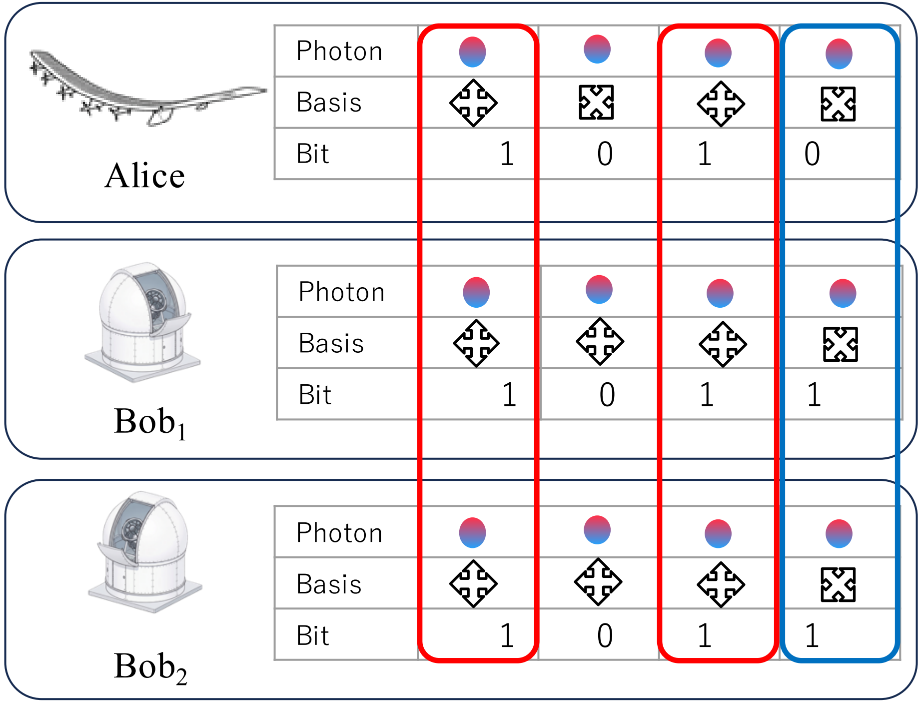
- This work proposes an HAP-assisted N-GHZ-QKD architecture over FSO for scalable group key distribution in 6G networks.
  1. Develop a Qiskit-based simulation framework incorporating major adverse effect in FSO
  2. Compare N-GHZ-QKD against conventional bipartite 2-GHZ-QKD
  3. Investigate the practical design and optimal setting of proposed system

Qiskit: An open-source SDK provided by IBM for quantum computing.

3-GHZ-QKD

$$|GHZ_3\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$



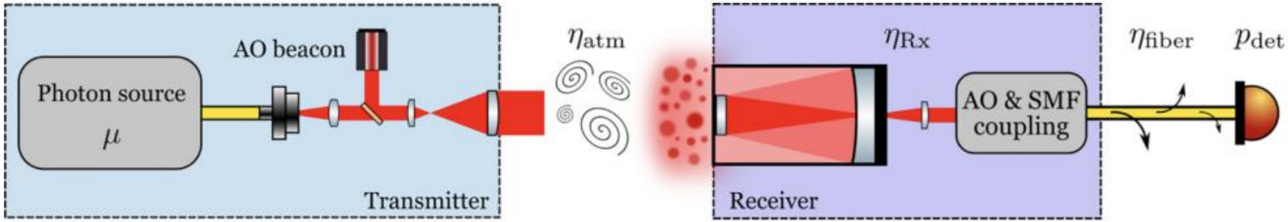


Record “11” as the sifted key

Adverse issues in FSO Channels

Photons are transmitted through FSO channel

$$\eta_{free\_space} = \eta_{atm} \times \eta_{R_X}$$
$$\eta_{R_X} = \eta_{DR_X} \times \eta_{SMF}$$



η_{atm}: Atmospheric Transmittance
η_{R_X} : The losses at the receiver
η_{DR_X}: The receiver collection efficiency
η_{SMF}: The fiber-coupling efficiency

These factors cause photon loss and photon state changes
→Affect key generation

System Model

HAP: Generates N-GHZ photon each time slot and retains one photon, distributes N-1 to ground users via FSO

Ground Users: Divided in to G subgroups, served sequentially with TDMA and each subgroup establishes a shared subgroup key

AKR: Asymptotic Key Rate

$$\mathrm{AKR}_g\,(P_\mathrm{Z}, N) = \mathcal{R}\langle Y_N^{(g)}\rangle r_\infty^{(g)} P_\mathrm{sift}(P_\mathrm{Z}, N)$$

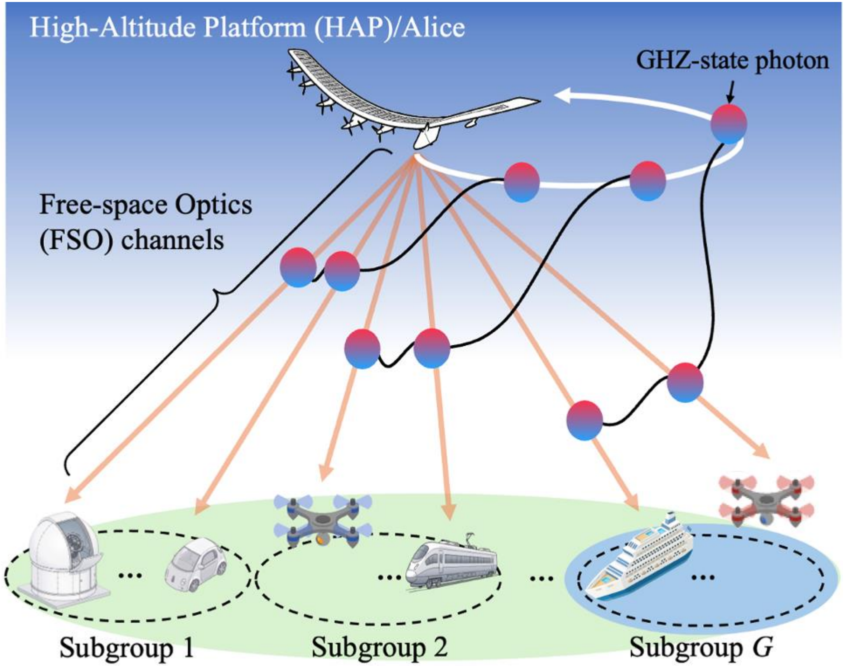
$\mathcal{R}$: Pulse repetition rate

$Y_N^{(g)}$: Probability of detection event

r_∞ : Secret-bit fraction per shared state

P_{sift} : Sifting probability

$$P_\mathrm{sift} = \begin{cases} P_\mathrm{Z}^2 + (1 - P_\mathrm{Z})^2 & \text{if } N = 2, \\ P_\mathrm{Z}^N & \text{if } N \geq 3, \end{cases}$$



Group Key Establishment

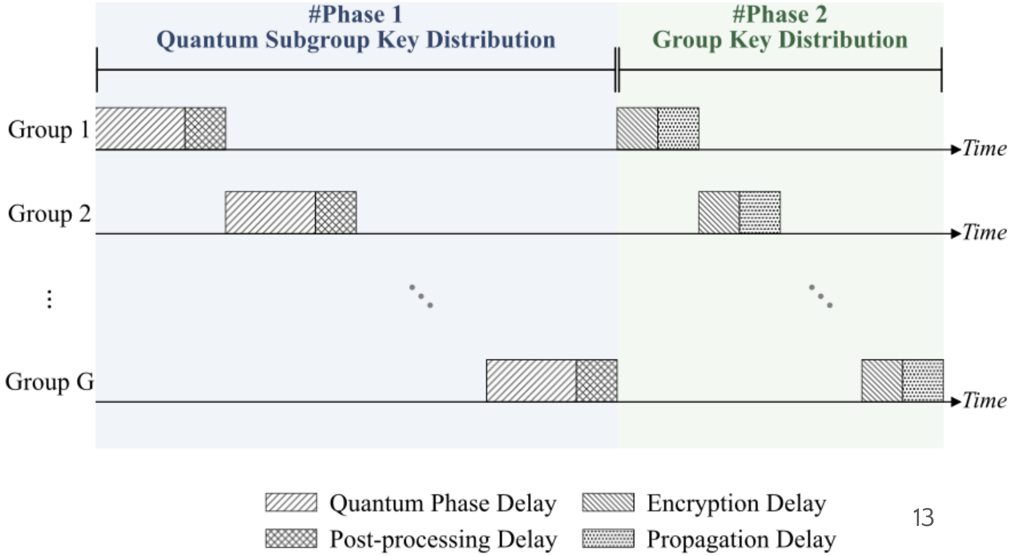
Phase 1: Quantum subgroup key distribution
- Quantum Phase Delay(Δt) : HAP distributes N-GHZ photons and each user measures
- Post-processing Delay(d_{pp}) : Announce basis, sift matching bits, estimate QBER

Phase 2:
- Encryption Delay(t_{enc}) : HAP generate group key and encrypt it
- Propagation Delay(d_{prop}) : HAP broadcasts group key to subgroup's user

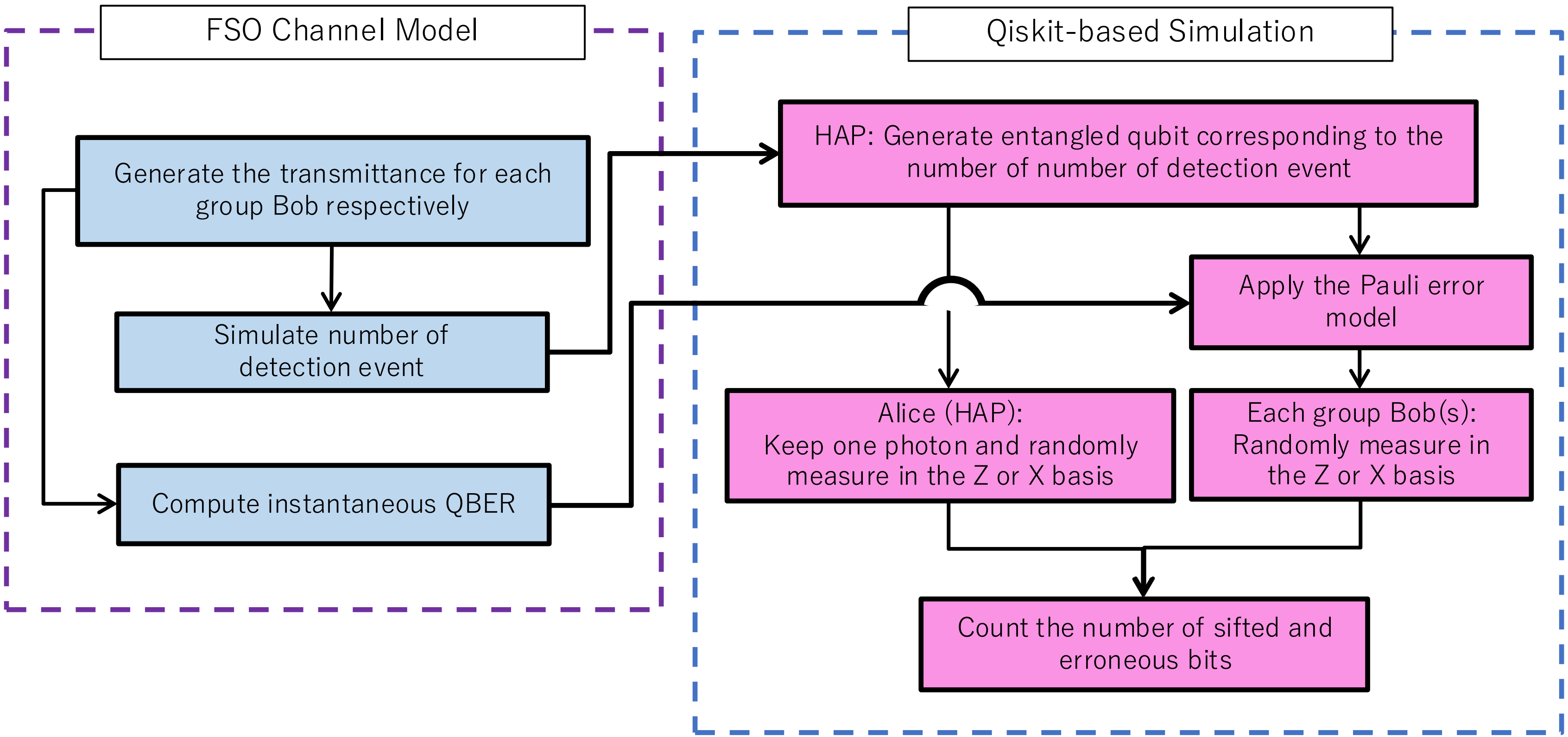
Secret Key Rate for All Group:

$$\mathrm{SKR_G}\left(P_Z, M, N\right)=\frac{\min\limits_{g\in\{1,\ldots,G\}}\left[\mathrm{AKR}_g\left(P_Z, N\right)\Delta t\right]}{G\left(\Delta t+d_{\mathrm{pp}}+t_{\mathrm{enc}}+d_{\mathrm{prop}}\right)}$$

$G=\left\lceil\frac{M}{N-1}\right\rceil$: Number of subgroups



Workflow of the Qiskit-based Simulation Framework per Time Slot



Result 1: Effective Transmittance and Z-basis Probability Dependence

Effective Transmittance η (Right Fig)

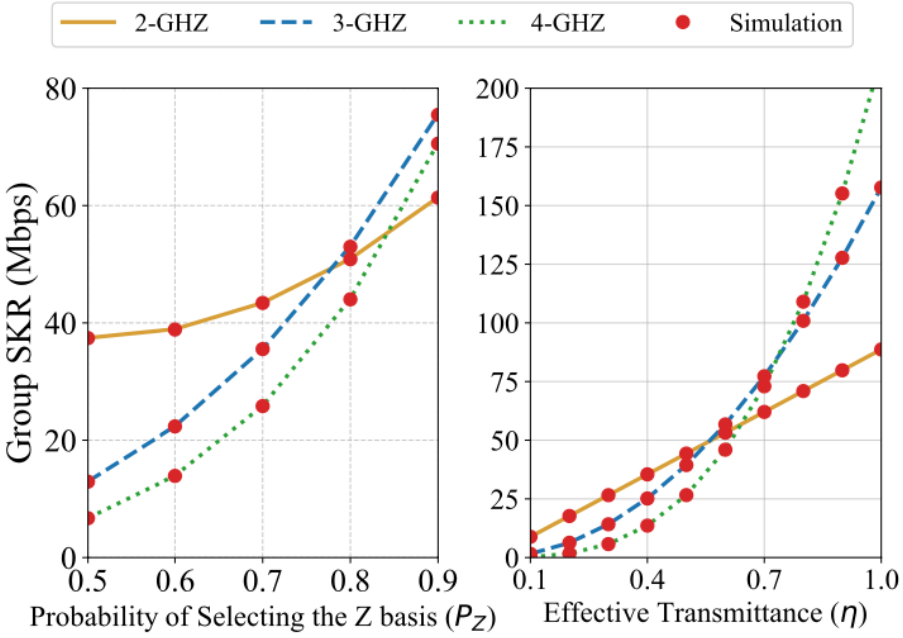
- 3/4-GHZ exceed 2-GHZ only at high η
- Driven by increased multi-user detection probability $Y_N^{(g)}$

Z-basis Probability P_Z (Left Fig)

- At $P_Z = 0.5$, 3-/4-GHZ underperform 2-GHZ
- P_{sift} scales as $P_Z^N (N \geq 3)$, favoring higher-order GHZ

Operating Regime

- Increasing P_Z to $0.8 - 0.9$ reverses the ordering
- Under the considered FSO conditions, this lets 3/4-GHZ achieve higher SKR



Result 2: Scalability with the Number of Users

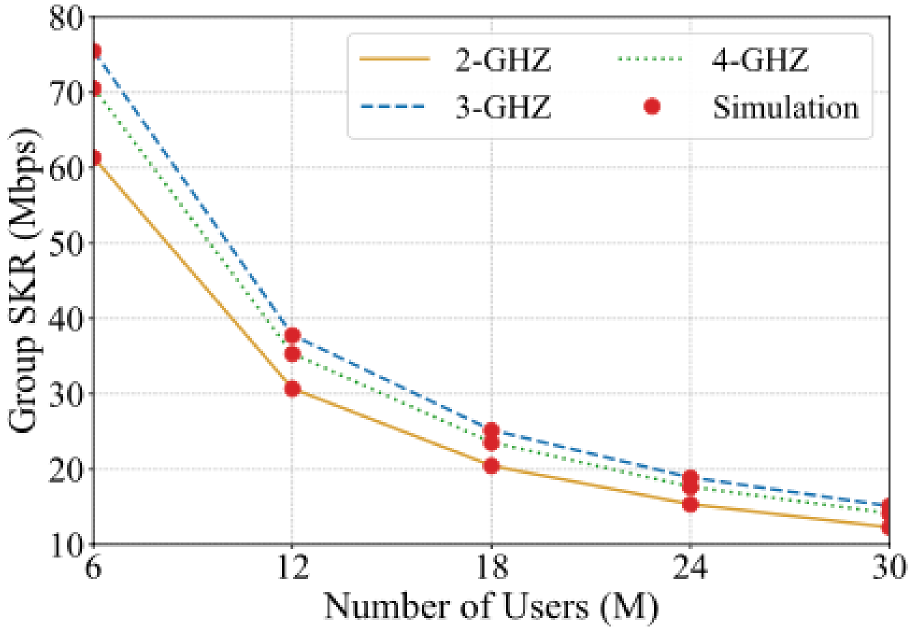
SKR Declines with M

- More users → more subgroups, $G=\lceil M/(N-1)\rceil$
- Longer TDMA reduces the overall group SKR

Higher-order GHZ Degrades More Slowly

- Larger N means fewer subgroups for the same M
- 3/4-GHZ sustain higher SKR as M grows

→E.g., for a required SKR of 20 Mbps, M is limited to 18 users



Result 2: Optimal Beam Waist

SKR Declines with M

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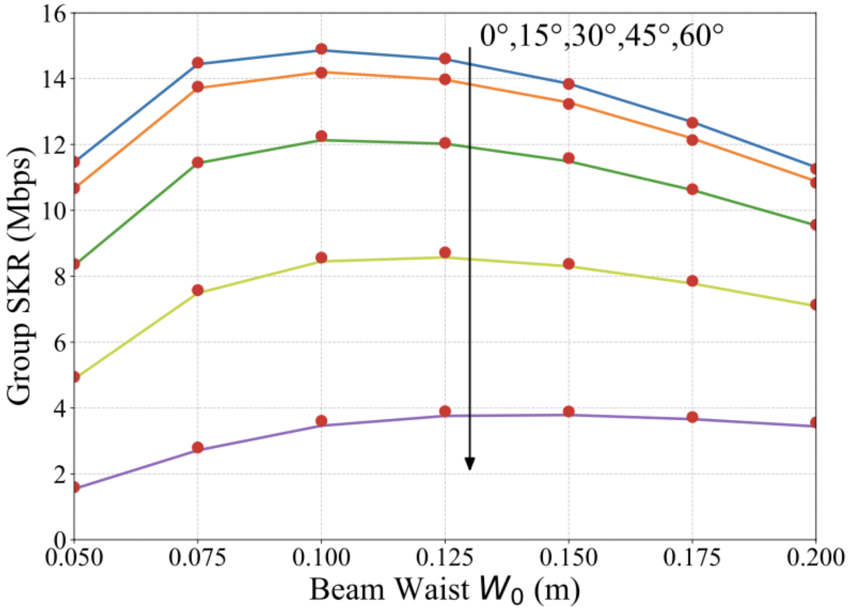


2. New Proposal

Motivation: Key Relaying in FSO/QKD System

Satellites are Used as Relays for Global Reach

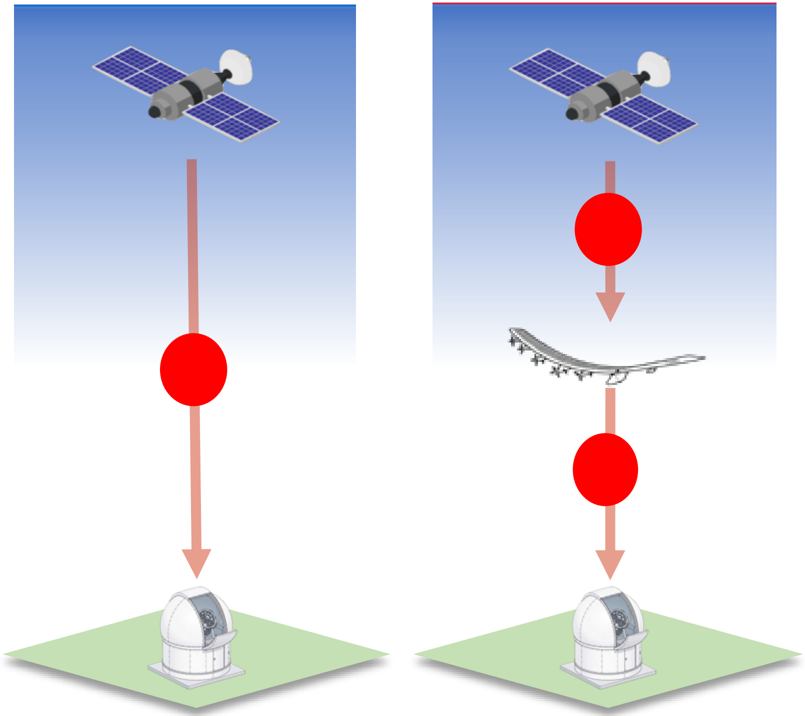
- For global-scale quantum communication, satellites act as relay nodes
- But the satellite-to-ground distance is itself very long

FSO Signals Weaken Over Distance

- Air absorbs, scatters, and disturbs light as it travels
- The longer the path through the atmosphere, the worse this gets

Relaying Solves the Photon Loss

- Each shorter link passes through less atmosphere, so less signal is lost



Motivation: Untrusted Relay Problem

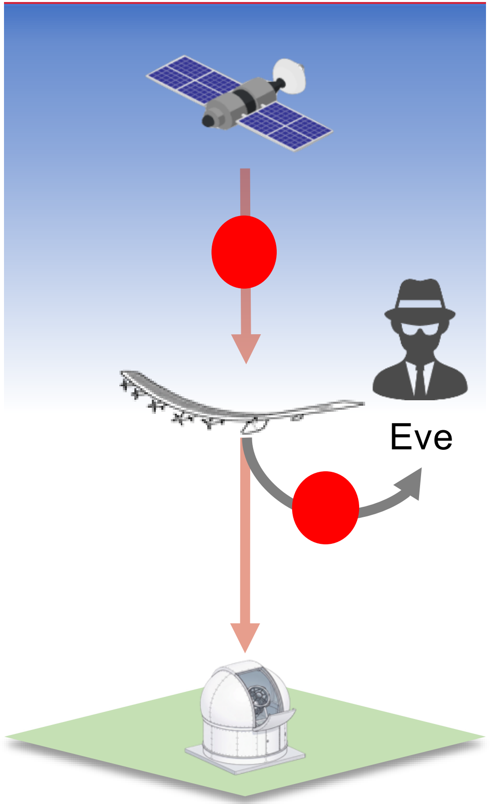
Relaying Assumes a Trusted Node

- HAP relay assumes the relay node is fully trustworthy
- In practice, a relay node may be compromised or partially untrusted

→ **Weakly Untrusted relay could access or leak the key it forwards**

Research Question:

How can key distribution remain secure through an weakly untrusted relay?



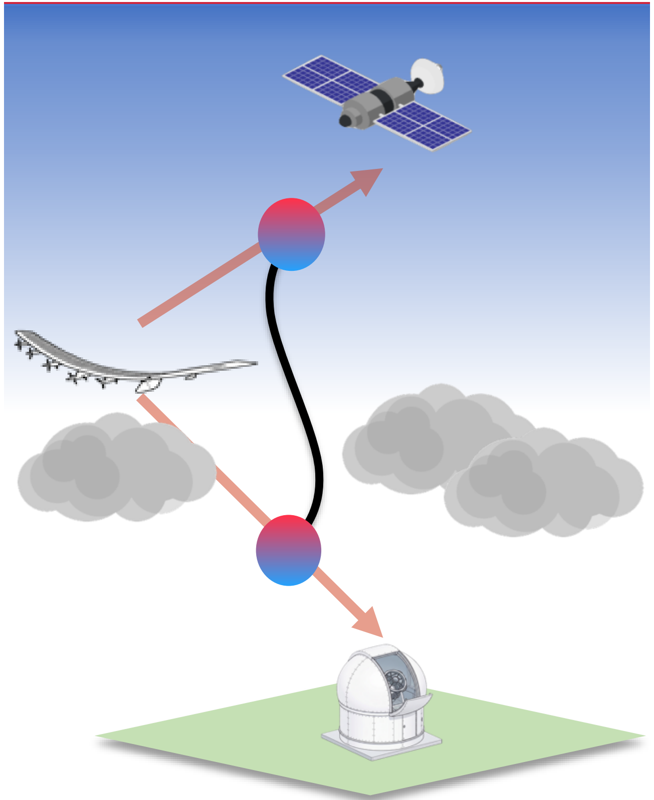
Possible Solution: Using BBM92 Protocol

How BBM92 Would Work

1. HAP generates an entangled photon pair and sends one photon to each side
2. Satellite and ground user must each detect their photon from the same pair
3. A shared key bit is only obtained when both detections succeed

Problem in this Architecture

- HAP-to-ground transmittance η is low due to cloud coverage
- Since the detection probability for each link is proportional to η, the simultaneous detection probability is η^2
- This makes the link from the HAP to the ground user a bottleneck



Proposed Solution: The Relay through Quantum Secret Sharing (QSS) Using 3-GHZ State

Proposed System

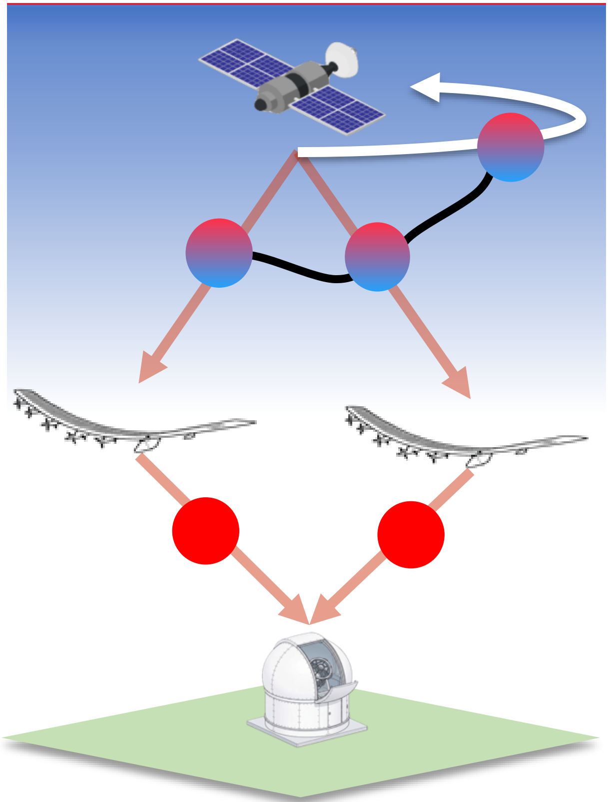
- Satellite, HAP1, and HAP2 generate a key seed by measuring the 3-GHZ state in either the X or Y basis and exchanging the basis information.
- Ground user receives information relayed from both HAPs.

Key Establish Phase

Phase 1: Satellite and HAPs derive key seed shares by GHZ measurement.

Phase 2: HAPs securely forward their shares to the ground user using the key generated by BB84 protocol.

- The satellite and the ground users obtain the key by combining the shared information from both HAPs.
- The key cannot be revealed using information from a single HAP.



Proposed Solution: 3-GHZ Quantum Secret Sharing

X/Y-Basis Patterns That Make GHZ-QSS Work

- K1, K2, and K3 represent the measurement results (Key seed), respectively.

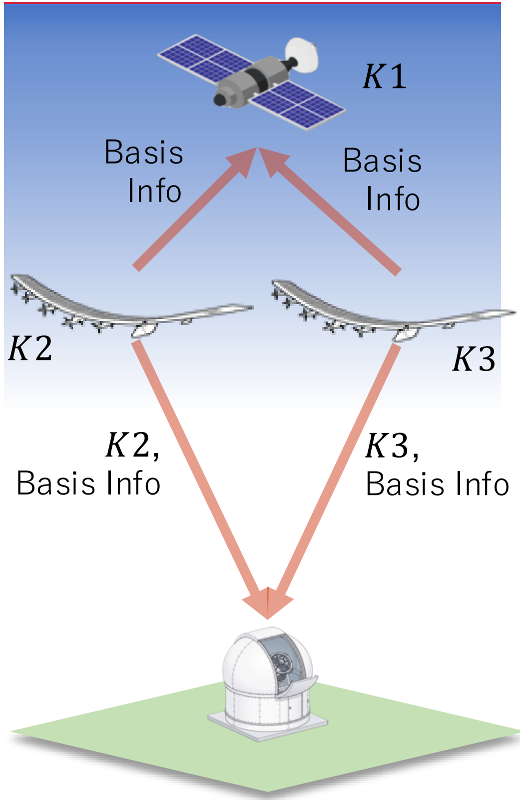
Example:

Basis Pattern	$K1$(Satellite)	$K2$(HAP1)	$K3$(HAP2)	Relation
XXX	0	1	1	$K1 = K2 \oplus K3$
XYY/YXY/YYX	0	0	1	$K1 = K2 \oplus K3 \oplus 1$

The probability of the above basis pattern appearing is 50%.

If the basis pattern (XXX, XYY/YXY/YYX) and the measurement results (K2, K3) are known from the above relationship, K1 can be derived.

→ K1 (the key) can be shared between the satellite and the ground user without the HAP becoming aware of K1.



Conclusion

Review of Previous Paper:

- HAP-assisted N-GHZ-QKD over FSO improves group SKR over bipartite QKD
- Simulation and theoretical result matched, confirming the validity of the developed framework

New Proposal :

- Identified the untrusted-relay in HAP-based key distribution
- Proposed a 3-GHZ QSS scheme:
  →No single HAP alone reveals the key